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Sensor element of NO.SUB.X .sensor and method for manufacturing sensor element of NO.SUB.X .sensor

Abstract

A sensor element includes: a base part containing an oxygen-ion conductive solid electrolyte as a constituent material; at least one internal space into which a measurement gas is introduced; and at least one pump cell including an internal electrode disposed to face the at least one internal space, an out-of-space pump electrode disposed at a location other than the at least one internal space, and a portion of the base part located between these electrodes, the internal electrode includes: an upper layer consisting of a noble metal, the solid electrolyte, and a pore; and a lower layer consisting of the noble metal and the solid electrolyte, and a volume ratio of the solid electrolyte in the lower layer is greater than a volume ratio of the solid electrolyte in the upper layer.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

(1) The present application claims priority from Japanese applications JP2021-008722, filed on Jan. 22, 2021 and JP2022-002294, filed on Jan. 11, 2022, the contents of which is hereby incorporated by reference into this application.

BACKGROUND OF THE INVENTION

Field of the Invention

(2) The present invention relates to a sensor element of a limiting current NO_x sensor, and, in particular, to a configuration of a measurement electrode of the sensor element.

Description of the Background Art

(3) As a device for measuring the concentration of NO_x in a measurement gas, such as a combustion gas and an exhaust gas from an internal combustion engine typified by an engine of a vehicle, a NO_x sensor including a sensor element including a base formed of oxygen-ion conductive solid electrolyte ceramics, such as zirconia (ZrO₂), has been known (see Japanese Patent No. 2885336 and WO 2019/188613, for example).

(4) The sensor element (a NO_x sensor element) of the NO_x sensor includes various electrodes (e.g., a pump electrode, a measurement electrode, and a reference electrode). These electrodes are porous cermet electrodes each formed of a composite material of a noble metal as a catalyst and zirconia as an electrolyte, and having a porous structure including many pores (cavities). As the catalytic noble metal, Pt and Pt with trace amounts of other substances (e.g., Rh and other noble metals) added are used. The NO_x sensor element utilizes, at operation thereof, a catalytic reaction of the catalytic noble metal used for the electrodes and oxygen-ion conductivity of zirconia used for the base, and is thus used in a state of being heated to a relatively high sensor element driving temperature (600° C. to 900° C.).

(5) The measurement gas, such as an exhaust gas from an engine, typically contains O₂. Thus, in a NO_x sensor as disclosed in Japanese Patent No. 2885336 and WO 2019/188613, O₂ in a measurement gas introduced into a NO_x sensor element is first removed (separated from NO_x) by an electrochemical pump cell of the NO_x sensor element, NO_x is then decomposed into O₂ and N₂ using a catalytic reaction of a measurement electrode, and a NO_x concentration of the measurement gas is measured based on the magnitude of an oxygen-ion current flowing through a measurement pump cell in the decomposition.

(6) When the measurement electrode is exposed to O₂, Pt and Pt/Rh of the electrode react with O₂ to generate oxides such as PtO, PtO₂, and RhO₂. When noble metals of the electrode decrease due to generation of the oxides, catalytic reactivity of the measurement electrode is reduced. These oxides have lower vapor pressure than Pt, and thus are likely to vaporize at a lower temperature than Pt. The vaporization leads to reduction in number of three-phase interfaces of a catalyst, an electrolyte, and a pore and two-phase interfaces of the catalyst and the electrolyte in the measurement electrode, and thus also causes reduction in catalytic reactivity of the measurement electrode.

(7) As described above, oxygen is removed in advance from the measurement gas reaching the measurement electrode, but a small amount of oxygen still remains in the measurement gas. Furthermore, O₂ generated by decomposition of NO_x might remain near the measurement electrode without promptly being ionized.

(8) Thus, by continued use of the NO_x sensor, NO_x sensitivity is gradually reduced due to reduction in catalytic reactivity of the measurement electrode.

(9) Elaborating on mass transfer near the measurement electrode at measurement, O.sub.2 and NOx in the measurement gas having reached the measurement electrode first transfer through pores of the measurement electrode (referred to as a mass transfer process). When O.sub.2 and NOx come into contact with a catalytic metal within the pores, Pt as a main component of the catalytic metal decomposes NO into N.sub.2 and O.sub.2, and further strips off an electron from O.sub.2 to generate an oxygen ion. The generated oxygen ion is taken, at a two-phase interface of the catalyst and the electrolyte, from Pt into zirconia as the solid electrolyte, and transfers within zirconia due to a potential difference caused between the measurement electrode and an external electrode by application of a voltage by an external power supply (referred to as a charge transfer process).

(10) If there are an oxygen ion not taken from the catalytic metal into zirconia and, further, oxygen before being ionized (generically referred to as surplus oxygen) in this process, however, the catalytic metal including Pt as the main component is oxidized by the surplus oxygen inside or at the surface of the electrode to generate PtO, PtO.sub.2, and the like. These oxides gradually vaporize.

(11) The inventors of the present invention have reasoned that reduction of the surplus oxygen not taken into zirconia can suppress vaporization of the oxides from the measurement electrode, and suppress reduction in sensitivity of the NOx sensor, and have conceived of the present invention through intense study.

(12) WO 2019/188613 notes that, at oxidation of Pt in the measurement electrode, impurities present in the electrode further reduce a temperature at which the oxides vaporize, and discloses that vaporization of the oxides is suppressed by providing a gettering layer gettering such impurities. WO 2019/188613, however, neither discloses nor suggests suppression of vaporization of the oxides by reduction of the surplus oxygen.

SUMMARY

(13) The present invention relates to a sensor element of a limiting current type NOx sensor, and is, in particular, directed to a configuration of a measurement electrode of the sensor element.

(14) According to the present invention, a sensor element of a limiting current type NOx sensor includes: a base part containing an oxygen-ion conductive solid electrolyte as a constituent material; at least one internal space into which a measurement gas is introduced; and at least one pump cell including an internal electrode disposed to face the at least one internal space, an out-of-space pump electrode disposed at a location other than the at least one internal space, and a portion of the base part located between the internal electrode and the out-of-space pump electrode, the internal electrode includes: an upper layer consisting of a noble metal, the solid electrolyte, and a pore; and a lower layer consisting of the noble metal and the solid electrolyte, and a volume ratio of the solid electrolyte in the lower layer is greater than a volume ratio of the solid electrolyte in the upper layer.

(15) A NOx sensor in which separation of the internal electrode is suitably suppressed even if the NOx sensor is used continuously is thereby achieved.

(16) It is preferable that the at least one internal space be a measurement internal space into which a measurement gas having an oxygen concentration adjusted in advance is introduced, the internal electrode be a measurement electrode disposed in the measurement internal space, and the at least one pump cell be a measurement pump cell including the measurement electrode, the out-of-space pump electrode, and a portion of the base part located between the measurement electrode and the out-of-space pump electrode.

(17) In this case, deterioration of the measurement electrode due to the presence of surplus oxygen not taken into the measurement electrode is suitably suppressed. A NOx sensor in which deterioration of measurement sensitivity is suitably suppressed even if the NOx sensor is used continuously is thereby achieved.

(18) It is thus an object of the present invention to provide a NOx sensor in which reduction in sensitivity due to oxidation of a measurement electrode is suppressed even if the NOx sensor is

used continuously.

(19) These and other objects, features, aspects and advantages of the present invention will become more apparent from the following detailed description of the present invention when taken in conjunction with the accompanying drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a diagram schematically showing one example of a configuration of a gas sensor **100**;
- (2) FIG. 2 is a model diagram showing a partial cross section of a measurement electrode **44** along a thickness direction of an element;
- (3) FIG. 3 is a model diagram of a partial cross section of a measurement electrode **44Z** having a configuration in which three phases coexist in the electrode as a whole;
- (4) FIG. 4 is a flowchart of processing at manufacture of a sensor element **101**;
- (5) FIG. 5 is a diagram showing procedures for forming a pattern to eventually be the measurement electrode **44**; and
- (6) FIG. 6 is a graph showing results of a continuous drive test.

DESCRIPTION OF THE PREFERRED EMBODIMENTS

(7) <General Configuration of Gas Sensor>

(8) FIG. 1 is a diagram schematically showing one example of a configuration of a gas sensor **100** according to an embodiment. The gas sensor **100** is a limiting current type NO_x sensor sensing NO_x and measuring the concentration thereof using a sensor element **101**. The gas sensor **100** further includes a controller **110** controlling operation of each part and identifying the NO_x concentration based on a NO_x current flowing through the sensor element **101**. FIG. 1 includes a vertical cross-sectional view taken along a longitudinal direction of the sensor element **101**.

(9) The sensor element **101** is a planar (elongated planar) element having a structure in which six solid electrolyte layers, namely, a first substrate layer **1**, a second substrate layer **2**, a third substrate layer **3**, a first solid electrolyte layer **4**, a spacer layer **5**, and a second solid electrolyte layer **6** each formed of zirconia (ZrO₂) (e.g., yttria stabilized zirconia (YSZ)) as an oxygen-ion conductive solid electrolyte are laminated in the stated order from a bottom side of FIG. 1. The solid electrolyte forming these six layers is dense and airtight. A surface on an upper side and a surface on a lower side of each of these six layers in FIG. 1 are hereinafter also simply referred to as an upper surface and a lower surface, respectively. A part of the sensor element **101** formed of the solid electrolyte as a whole is generically referred to as a base part.

(10) The sensor element **101** is manufactured, for example, by performing predetermined processing, printing of circuit patterns, and the like on ceramic green sheets corresponding to the respective layers, then laminating them, and further firing them for integration.

(11) Between a lower surface of the second solid electrolyte layer **6** and an upper surface of the first solid electrolyte layer **4** in one leading end portion of the sensor element **101**, a first diffusion control part **11** doubling as a gas inlet **10**, a buffer space **12**, a second diffusion control part **13**, a first internal space **20**, a third diffusion control part **30**, a second internal space **40**, a fourth diffusion control part **60**, and a third internal space **61** are formed adjacent to each other to communicate in the stated order.

(12) The buffer space **12**, the first internal space **20**, the second internal space **40**, and the third internal space **61** are spaces (regions) inside the sensor element **101** looking as if they were provided by hollowing out the spacer layer **5**, and having an upper portion, a lower portion, and a side portion respectively defined by the lower surface of the second solid electrolyte layer **6**, the upper surface of the first solid electrolyte layer **4**, and a side surface of the spacer layer **5**. The gas inlet **10** may similarly look as if it was provided by hollowing out the spacer layer **5** at a leading

end surface (at the left end in FIG. 1) of the sensor element **101** separately from the first diffusion control part **11**. In this case, the first diffusion control part **11** is formed inside and adjacent to the gas inlet **10**.

(13) The first diffusion control part **11**, the second diffusion control part **13**, the third diffusion control part **30**, and the fourth diffusion control part **60** are each provided as two horizontally long slits (whose openings have longitudinal directions perpendicular to the page of FIG. 1). A part extending from the gas inlet **10** to the third internal space **61** is also referred to as a gas distribution part.

(14) At a location farther from the leading end than the gas distribution part is, a reference gas introduction space **43** having a side portion defined by a side surface of the first solid electrolyte layer **4** is provided between an upper surface of the third substrate layer **3** and a lower surface of the spacer layer **5**. For example, air is introduced into the reference gas introduction space **43** as a reference gas at measurement of the NO_x concentration.

(15) An air introduction layer **48** is a layer formed of porous alumina, and the reference gas is introduced into the air introduction layer **48** through the reference gas introduction space **43**. The air introduction layer **48** is formed to cover a reference electrode **42**.

(16) The reference electrode **42** is an electrode formed to be sandwiched between the upper surface of the third substrate layer **3** and the first solid electrolyte layer **4**, and the air introduction layer **48** leading to the reference gas introduction space **43** is provided around the reference electrode **42** as described above. As will be described below, an oxygen concentration (oxygen partial pressure) in the first internal space **20** and the second internal space **40** can be measured using the reference electrode **42**.

(17) In the gas distribution part, the gas inlet **10** (first diffusion control part **11**) is a part opening to an external space, and a measurement gas is taken from the external space into the sensor element **101** through the gas inlet **10**.

(18) The first diffusion control part **11** is a part providing predetermined diffusion resistance to the taken measurement gas.

(19) The buffer space **12** is a space provided to guide the measurement gas introduced through the first diffusion control part **11** to the second diffusion control part **13**.

(20) The second diffusion control part **13** is a part providing predetermined diffusion resistance to the measurement gas introduced from the buffer space **12** into the first internal space **20**.

(21) In introducing the measurement gas from outside the sensor element **101** into the first internal space **20**, the measurement gas having abruptly been taken into the sensor element **101** through the gas inlet **10** due to pressure fluctuations (pulsation of exhaust pressure in a case where the measurement gas is an exhaust gas of a vehicle) of the measurement gas in the external space is not directly introduced into the first internal space **20** but is introduced into the first internal space **20** after concentration fluctuations of the measurement gas are canceled through the first diffusion control part **11**, the buffer space **12**, and the second diffusion control part **13**. This makes the concentration fluctuations of the measurement gas introduced into the first internal space **20** almost negligible.

(22) The first internal space **20** is provided as a space to adjust oxygen partial pressure of the measurement gas introduced through the second diffusion control part **13**. The oxygen partial pressure is adjusted by operation of a main pump cell **21**.

(23) The main pump cell **21** is an electrochemical pump cell including an inner pump electrode **22**, an outer (out-of-space) pump electrode **23**, and the second solid electrolyte layer **6** sandwiched between these electrodes. The inner pump electrode **22** has a ceiling electrode portion **22a** provided on substantially the entire lower surface of a portion of the second solid electrolyte layer **6** facing the first internal space **20**, and the outer pump electrode **23** is provided in a region, on an upper surface of the second solid electrolyte layer **6** (one main surface of the sensor element **101**), corresponding to the ceiling electrode portion **22a** to be exposed to the external space.

(24) The inner pump electrode **22** is formed on upper and lower solid electrolyte layers (the second solid electrolyte layer **6** and the first solid electrolyte layer **4**) defining the first internal space **20**. Specifically, the ceiling electrode portion **22a** is formed on the lower surface of the second solid electrolyte layer **6**, which provides a ceiling surface to the first internal space **20**, and a bottom electrode portion **22b** is formed on the upper surface of the first solid electrolyte layer **4**, which provides a bottom surface to the first internal space **20**. The ceiling electrode portion **22a** and the bottom electrode portion **22b** are connected by a conducting portion (not illustrated) provided on a side wall surface (an inner surface) of the spacer layer **5** forming opposite side wall portions of the first internal space **20**.

(25) The ceiling electrode portion **22a** and the bottom electrode portion **22b** are provided to be rectangular in plan view. Only the ceiling electrode portion **22a** or only the bottom electrode portion **22b** may be provided.

(26) The inner pump electrode **22** and the outer pump electrode **23** are each formed as a porous cermet electrode. In particular, the inner pump electrode **22** to be in contact with the measurement gas is formed using a material having a weakened reducing ability with respect to a NO_x component in the measurement gas. For example, the inner pump electrode **22** is formed as a cermet electrode of an Au—Pt alloy containing Au of approximately 0.6 wt % to 1.4 wt % and ZrO₂ to have a porosity of 5% to 40% and a thickness of 5 μm to 20 μm. A weight ratio Pt:ZrO₂ of the Au—Pt alloy and ZrO₂ is only required to be approximately 7.0:3.0 to 5.0:5.0.

(27) On the other hand, the outer pump electrode **23** is formed, for example, as a cermet electrode of Pt or an alloy thereof and ZrO₂ to be rectangular in plan view.

(28) The main pump cell **21** can pump out oxygen in the first internal space **20** to the external space or pump in oxygen in the external space to the first internal space **20** by applying, from a variable power supply **24**, a desired pump voltage V_{p0} between the inner pump electrode **22** and the outer pump electrode **23** to allow a main pump current I_{p0} to flow between the inner pump electrode **22** and the outer pump electrode **23** in a positive or negative direction. The pump voltage V_{p0} applied between the inner pump electrode **22** and the outer pump electrode **23** in the main pump cell **21** is also referred to as a main pump voltage V_{p0} .

(29) To detect the oxygen concentration (oxygen partial pressure) in an atmosphere in the first internal space **20**, the inner pump electrode **22**, the second solid electrolyte layer **6**, the spacer layer **5**, the first solid electrolyte layer **4**, the third substrate layer **3**, and the reference electrode **42** constitute a main sensor cell **80** as an electrochemical sensor cell.

(30) The oxygen concentration (oxygen partial pressure) in the first internal space **20** can be known by measuring electromotive force V_0 in the main sensor cell **80**.

(31) Furthermore, the controller **110** performs feedback control of the main pump voltage V_{p0} so that the electromotive force V_0 is constant, thereby to control the main pump current I_{p0} . The oxygen concentration in the first internal space **20** is thereby maintained at a predetermined constant value.

(32) The third diffusion control part **30** is a part providing predetermined diffusion resistance to the measurement gas having an oxygen concentration (oxygen partial pressure) controlled by operation of the main pump cell **21** in the first internal space **20**, and guiding the measurement gas to the second internal space **40**.

(33) The second internal space **40** is provided as a space to further adjust the oxygen partial pressure of the measurement gas introduced through the third diffusion control part **30**. The oxygen partial pressure is adjusted by operation of an auxiliary pump cell **50**. The oxygen concentration of the measurement gas is adjusted with higher accuracy in the second internal space **40**.

(34) After the oxygen concentration (oxygen partial pressure) is adjusted in advance in the first internal space **20**, the auxiliary pump cell **50** further adjusts the oxygen partial pressure of the measurement gas introduced through the third diffusion control part **30** in the second internal space

40.

(35) The auxiliary pump cell **50** is an auxiliary electrochemical pump cell including an auxiliary pump electrode **51**, the outer pump electrode **23** (not limited to the outer pump electrode **23** and only required to be any appropriate electrode outside the sensor element **101**), and the second solid electrolyte layer **6**. The auxiliary pump electrode **51** has a ceiling electrode portion **51a** provided on substantially the entire lower surface of a portion of the second solid electrolyte layer **6** facing the second internal space **40**.

(36) The auxiliary pump electrode **51** is provided in the second internal space **40** in a similar form to the inner pump electrode **22** provided in the first internal space **20** described previously. That is to say, the ceiling electrode portion **51a** is formed on the second solid electrolyte layer **6**, which provides a ceiling surface to the second internal space **40**, and a bottom electrode portion **51b** is formed on the first solid electrolyte layer **4**, which provides a bottom surface to the second internal space **40**. The ceiling electrode portion **51a** and the bottom electrode portion **51b** are rectangular in plan view, and are connected by a conducting portion (not illustrated) provided on the side wall surface (inner surface) of the spacer layer **5** forming opposite side wall portions of the second internal space **40**.

(37) As with the inner pump electrode **22**, the auxiliary pump electrode **51** is formed using a material having a weakened reducing ability with respect to the NO_x component in the measurement gas.

(38) The auxiliary pump cell **50** can pump out oxygen in an atmosphere in the second internal space **40** to the external space or pump in oxygen in the external space to the second internal space **40** by applying a desired voltage (an auxiliary pump voltage) V_{p1} between the auxiliary pump electrode **51** and the outer pump electrode **23** under control performed by the controller **110**.

(39) To control the oxygen partial pressure in the atmosphere in the second internal space **40**, the auxiliary pump electrode **51**, the reference electrode **42**, the second solid electrolyte layer **6**, the spacer layer **5**, the first solid electrolyte layer **4**, and the third substrate layer **3** constitute an auxiliary sensor cell **81** as an electrochemical sensor cell.

(40) The auxiliary pump cell **50** performs pumping using a variable power supply **52** whose voltage is controlled based on electromotive force V_1 detected by the auxiliary sensor cell **81** in accordance with the oxygen partial pressure in the second internal space **40**. The oxygen partial pressure in the atmosphere in the second internal space **40** is thereby controlled to a low partial pressure having substantially no effect on measurement of NO_x.

(41) At the same time, a resulting auxiliary pump current I_{p1} is used to control the electromotive force in the main sensor cell **80**. Specifically, the auxiliary pump current I_{p1} is input, as a control signal, into the main sensor cell **80**, and, through control of the electromotive force V_0 therein, the oxygen partial pressure of the measurement gas introduced through the third diffusion control part **30** into the second internal space **40** is controlled to have a gradient that is always constant. In use as the NO_x sensor, the oxygen concentration in the second internal space **40** is maintained at a constant value of approximately 0.001 ppm by the action of the main pump cell **21** and the auxiliary pump cell **50**.

(42) The fourth diffusion control part **60** is a part providing predetermined diffusion resistance to the measurement gas having an oxygen concentration (oxygen partial pressure) controlled by operation of the auxiliary pump cell **50** in the second internal space **40**, and guiding the measurement gas to the third internal space **61**.

(43) The third internal space **61** is provided as a space (measurement internal space) to perform processing concerning measurement of the nitrogen oxide (NO_x) concentration of the measurement gas introduced through the fourth diffusion control part **60**. The NO_x concentration is measured by operation of a measurement pump cell **41** in the third internal space **61**. The measurement gas having the oxygen concentration adjusted with high accuracy in the second internal space **40** is introduced into the third internal space **61**, so that the NO_x concentration can be measured with

high accuracy in the gas sensor **100**.

(44) The measurement pump cell **41** measures the NO_x concentration of the measurement gas in the third internal space **61**. The measurement pump cell **41** is an electrochemical pump cell including a measurement electrode **44**, the outer pump electrode **23**, the second solid electrolyte layer **6**, the spacer layer **5**, and the first solid electrolyte layer **4**. The measurement electrode **44** is provided on an upper surface of a portion of the first solid electrolyte layer **4** facing the third internal space **61** to be separated from the third diffusion control part **30**.

(45) The measurement electrode **44** is a porous cermet electrode of a noble metal and a solid electrolyte. For example, the measurement electrode **44** is formed as a cermet electrode of Pt or an alloy of Pt and another noble metal, such as Rh, and ZrO₂ as a constituent material for the sensor element **101**. The measurement electrode **44** also functions as a NO_x reduction catalyst to reduce NO_x present in the atmosphere in the third internal space **61**.

(46) The measurement pump cell **41** can pump out oxygen generated through decomposition of NO_x in an atmosphere around the measurement electrode **44**, and detect the amount of generated oxygen as a pump current I_{p2} under control performed by the controller **110**.

(47) In the present embodiment, the measurement electrode **44** has a two-layer configuration including a region where pores are not present and a region where pores are present. Details thereof will be described below.

(48) To detect the oxygen partial pressure around the measurement electrode **44**, the second solid electrolyte layer **6**, the spacer layer **5**, the first solid electrolyte layer **4**, the third substrate layer **3**, the measurement electrode **44**, and the reference electrode **42** constitute a measurement sensor cell **82** as an electrochemical sensor cell. A variable power supply **46** is controlled based on electromotive force V_2 detected by the measurement sensor cell **82** in accordance with the oxygen partial pressure around the measurement electrode **44**.

(49) NO_x in the measurement gas introduced into the third internal space **61** is reduced by the measurement electrode **44** ($2\text{NO} + 2\text{e}^- + \text{O}_2 \rightarrow 2\text{NO}_2$) to generate oxygen. Oxygen as generated is to be pumped by the measurement pump cell **41**, and, in this case, a voltage (measurement pump voltage) V_{p2} of the variable power supply **46** is controlled so that the electromotive force V_2 detected by the measurement sensor cell **82** is constant. The amount of oxygen generated around the measurement electrode **44** is proportional to the NO_x concentration of the measurement gas, and thus the NO_x concentration of the measurement gas is to be calculated using the pump current I_{p2} in the measurement pump cell **41**. The pump current I_{p2} is hereinafter also referred to as a NO_x current I_{p2} .

(50) In the case that the measurement electrode **44**, the first solid electrolyte layer **4**, the third substrate layer **3**, and the reference electrode **42** are combined to constitute an oxygen partial pressure detection means as an electrochemical sensor cell, electromotive force in accordance with a difference between the amount of oxygen generated through reduction of a NO_x component in the atmosphere around the measurement electrode **44** and the amount of oxygen contained in reference air can be detected, and the concentration of the NO_x component in the measurement gas can thereby be determined.

(51) The second solid electrolyte layer **6**, the spacer layer **5**, the first solid electrolyte layer **4**, the third substrate layer **3**, the outer pump electrode **23**, and the reference electrode **42** constitute an electrochemical sensor cell **83**, and oxygen partial pressure of the measurement gas outside the sensor can be detected using electromotive force V_{ref} obtained by the sensor cell **83**.

(52) The sensor element **101** further includes a heater part **70** playing a role in temperature adjustment of heating the sensor element **101** and maintaining the temperature thereof to enhance oxygen ion conductivity of the solid electrolyte forming the base part.

(53) The heater part **70** mainly includes a heater electrode **71**, a heater element **72**, a heater lead **72a**, a through hole **73**, a heater insulating layer **74**, and a heater resistance detection lead, which is not illustrated in FIG. 1. A portion of the heater part **70** other than the heater electrode **71** is buried

in the base part of the sensor element **101**.

(54) The heater electrode **71** is an electrode formed to be in contact with a lower surface of the first substrate layer **1** (the other main surface of the sensor element **101**).

(55) The heater element **72** is a resistive heating element provided between the second substrate layer **2** and the third substrate layer **3**. The heater element **72** generates heat by being powered from a heater power supply, which is not illustrated in FIG. **1**, outside the sensor element **101** through the heater electrode **71**, the through hole **73**, and the heater lead **72a**, which constitute a current-carrying path. The heater element **72** is formed of Pt, or contains Pt as a main component. The heater element **72** is buried, in a predetermined range of the sensor element **101** in which the gas distribution part is provided, to oppose the gas distribution part in a thickness direction of the element. The heater element **72** is provided to have a thickness of approximately 10 μm to 20 μm .

(56) In the sensor element **101**, each part of the sensor element **101** can be heated to a predetermined temperature and the temperature can be maintained by allowing a current to flow through the heater electrode **71** to the heater element **72** to thereby cause the heater element **72** to generate heat. Specifically, the sensor element **101** is heated so that the temperature of the solid electrolyte and the electrodes in the vicinity of the gas distribution part is approximately 700° C. to 900° C. The oxygen ion conductivity of the solid electrolyte forming the base part of the sensor element **101** is enhanced by the heating. A heating temperature of the heater element **72** when the gas sensor **100** is in use (when the sensor element **101** is driven) is referred to as a sensor element driving temperature.

(57) A degree of heat generation (heater temperature) of the heater element **72** is grasped by the magnitude of a resistance value (heater resistance) of the heater element **72**.

(58) Although not illustrated in FIG. **1**, a thermal shock resistant protective layer as a single- or multi-porous layer covering the sensor element **101** may further be provided outside in a predetermined range on a side of the one leading end portion (side of the left end in FIG. **1**) of the sensor element **101**. The thermal shock resistant protective layer is provided to prevent cracking of the sensor element **101** due to thermal shock caused by moisture contained in the measurement gas adhering to the sensor element **101** and condensing when the gas sensor **100** is in use, and prevent poisoning substances coexisting in the measurement gas from entering into the sensor element **101**. A laminar gap (gap layer) may be formed between the sensor element **101** and the thermal shock resistant protective layer.

(59) <Detailed Configuration of Measurement Electrode>

(60) A configuration of the measurement electrode **44** of the sensor element **101** having a configuration as described above will be described in more detail next. FIG. **2** is a model diagram showing a partial cross section of the measurement electrode **44** along the thickness direction of the element. Dashed lines La and Lb shown in FIG. **2** are added for ease of understanding.

(61) In FIG. **2**, a portion below the dashed line La is the first solid electrolyte layer **4** formed of a solid electrolyte (zirconia), and the measurement electrode **44** is provided on the first solid electrolyte layer **4**. A portion above the measurement electrode **44** is the third internal space **61** (more particularly, a portion of the third internal space **61** not occupied by the measurement electrode **44**, but hereinafter simply referred to as the third internal space **61** for the sake of convenience).

(62) As described above, the measurement electrode **44** is the porous cermet electrode of the noble metal, such as Pt and an alloy of Pt and Rh and the like, and the solid electrolyte (zirconia). Thus, as shown in FIG. **2**, the measurement electrode **44** has a configuration that there coexist a portion formed of a noble metal NM of white in FIG. **2**, a portion formed of a solid electrolyte SE of gray (or light gray) in FIG. **2**, and a portion formed of a pore (cavity) CV of black (or dark gray) in FIG. **2**. As with the solid electrolyte SE, the first solid electrolyte layer **4** is of gray in FIG. **2**. This is because they are both formed of zirconia. The third internal space **61** is of black in FIG. **2** as with the pore CV, and this is because they are both spaces or gaps.

(63) The measurement electrode **44** according to the present embodiment, however, does not have a configuration in which the three portions (phases) formed of the noble metal NM, the solid electrolyte SE, and the pore CV randomly coexist as a whole, but has a two-layer configuration including a lower layer (also referred to as a two-phase region) **44L** where only the noble metal NM and the solid electrolyte SE randomly coexist, and the pore CV is not present and an upper layer (also referred to as a three-phase region) **44U** where the noble metal NM, the solid electrolyte SE, and the pore CV randomly coexist. A boundary between the lower layer **44L** and the upper layer **44U** is shown in the dashed line Lb in FIG. 2.

(64) Viewed another way, the dashed line La shows a lowermost end of a range in which the noble metal NM is present in a thickness direction of the measurement electrode **44**, and the dashed line Lb shows a lowermost end of a range in which the pore CV is present in the thickness direction of the measurement electrode **44**. Expressed yet another way, it can be said that the dashed line La is a boundary between a region where the noble metal NM is present and a region where the noble metal NM is not present in the thickness direction of the measurement electrode **44**, and the dashed line Lb is a boundary between a region where the pore CV is present and a region where the pore CV is not present in the thickness direction of the measurement electrode **44**.

(65) A portion of the solid electrolyte SE forming the measurement electrode **44** contiguous with the base part formed of the solid electrolyte of the sensor element **101** (a portion contiguous with the base part beyond the dashed line La) is hereinafter particularly referred to as a contiguous region RE. Furthermore, a portion of the contiguous region RE belonging to the lower layer **44L** (a portion located between the dashed line La and the dashed line Lb) is referred to as a lower contiguous region REL, and a portion of the contiguous region RE belonging to the upper layer **44U** (a portion located between the dashed line Lb and the third internal space **61**) is referred to as an upper contiguous region REU.

(66) Changes in the thickness direction of average volume ratios of respective portions of the measurement electrode **44** are further shown to the left of the model diagram in FIG. 2. Herein, t_0 , $t_{\text{sub.L}}$, and $t_{\text{sub.U}}$ are respectively the thickness of the measurement electrode **44** as a whole, the thickness of the lower layer **44L**, and the thickness of the upper layer **44U**.

(67) Although varying locally, the volume ratios of the noble metal NM, the solid electrolyte SE, and the pore CV in the upper layer **44U** are, on average, substantially constant in the entire range of the thickness $t_{\text{sub.U}}$ of the upper layer **44U** as shown in FIG. 2. In contrast, in the lower layer **44L**, the volume ratio of the pore CV is zero, while the volume ratio of the solid electrolyte SE is substantially constant in the entire range of the thickness, and has a greater value than in the upper layer **44U**. The volume ratio of the noble metal NM is constant in the measurement electrode **44** in FIG. 2, but may not necessarily be constant.

(68) That is to say, the measurement electrode **44** of the sensor element **101** according to the present embodiment is characterized in that the measurement electrode **44** as a whole is provided as the porous cermet electrode in which the noble metal NM, the solid electrolyte SE, and the pore CV coexist, but these three phases actually coexist only in the upper layer **44U** provided on a side of an upper surface of the electrode to have the predetermined thickness $t_{\text{sub.U}}$, and a range below the upper layer **44U** having the predetermined thickness $t_{\text{sub.L}}$ from the upper surface of the first solid electrolyte layer **4** is the lower layer **44L** having a two-phase configuration in which the pore CV is not present, and thus the volume ratio of the solid electrolyte SE is greater than that in the upper layer **44U**.

(69) In other words, in the measurement electrode **44**, the boundary (dashed line La) between the base part formed of the solid electrolyte (more particularly, the first solid electrolyte layer **4**) and the measurement electrode **44** is different from the boundary (dashed line Lb) between the region where the pore CV is present and the region where the pore CV is not present.

(70) FIG. 3 is a model diagram of a partial cross section of a measurement electrode **44Z** having a configuration in which three phases randomly coexist in the electrode as a whole shown for

comparison. A location at which the measurement electrode **44Z** is disposed in the sensor element is the same as a location at which the measurement electrode **44** is disposed in the sensor element. A portion of white in FIG. **3** is the portion formed of the noble metal NM, a portion of gray in FIG. **3** is the portion formed of the solid electrolyte SE, and a portion of black in FIG. **3** is the portion formed of the pore CV also in the measurement electrode **44Z** shown in FIG. **3**.

(71) Changes in the thickness direction of average volume ratios of respective portions of the measurement electrode **44Z** are also shown to the left of the model diagram in FIG. **3**. The measurement electrode **44Z**, however, has the same thickness t_0 as the measurement electrode **44**, and the average volume ratios of the respective phases are the same as those in the upper layer **44U** of the measurement electrode **44**.

(72) Although varying locally, the volume ratios of the noble metal NM, the solid electrolyte SE, and the pore CV in the measurement electrode **44Z** are, on average, substantially constant in the entire range of the thickness t_0 of the measurement electrode **44Z** as shown in FIG. **3**.

(73) In the measurement electrode **44Z**, the boundary between the region where the pore CV is present and the region where the pore CV is not present matches the boundary between the base part formed of the solid electrolyte (more particularly, the first solid electrolyte layer **4**) and the measurement electrode **44Z**.

(74) The measurement electrode **44** and the measurement electrode **44Z** have a mass transfer process in the electrode in common. That is to say, NO_x and trace amounts of $\text{O}_{\text{sub.2}}$, which are contained in the measurement gas having reached the measurement electrode **44** or the measurement electrode **44Z** disposed in the third internal space **61** from outside the element, transfer through the pore CV, and come into contact with the noble metal NM as a catalyst. Pt as a main component of the noble metal NM causes decomposition of NO_x to generate $\text{O}_{\text{sub.2}}$, and strips off an electron from $\text{O}_{\text{sub.2}}$ to generate an oxygen ion. The generated oxygen ion is taken, at a two-phase interface of the noble metal NM and the solid electrolyte SE, from Pt into the solid electrolyte SE, and transfers within the sensor element **101** due to a potential difference caused between the measurement electrode **44** or the measurement electrode **44Z** and the outer pump electrode **23** in the measurement pump cell **41**.

(75) In a case of the measurement electrode **44Z** having a three-phase configuration as a whole, however, the pore CV is present near or adjacent to the first solid electrolyte layer **4**, and an interface of the solid electrolyte SE and the pore CV is present at a relatively high ratio, whereas an interface of the noble metal NM and the solid electrolyte SE is present at a relatively low ratio. Thus, surplus oxygen coming into contact with the noble metal but not taken into the solid electrolyte is likely to be generated, and thus deterioration of the measurement electrode caused by oxidation of the noble metal containing Pt as the main component by the surplus oxygen is likely to occur.

(76) In contrast, in the case of the measurement electrode **44**, while the pore CV through which oxygen is distributed is present only in the upper layer **44U**, the solid electrolyte SE present in the lower layer **44L** is almost in the lower contiguous region REL contiguous with the base part, and is only in contact with the noble metal NM. Thus, a ratio of the two-phase interface of the noble metal NM and the solid electrolyte SE is greater than that in a conventional gas sensor, and, in particular, only the two-phase interface is present in the lower layer **44L**, so that the surplus oxygen is less likely to accumulate in the pore CV to be in contact with the noble metal NM, and deteriorate the measurement electrode **44**.

(77) That is to say, in the gas sensor **100** according to the present embodiment, deterioration of the measurement electrode **44** due to the presence of the surplus oxygen is suitably suppressed. Deterioration of measurement sensitivity is thus suitably suppressed even if the gas sensor **100** is used continuously.

(78) It is preferable that the measurement electrode **44** have a thickness of 10 μm to 30 μm , and a thickness ratio $t_{\text{sub.U}}:t_{\text{sub.L}}$ be in a range of 95:5 to 10:90. In other words, a ratio of a distance to

from an uppermost portion of the measurement electrode **44** to the lowermost end (dashed line Lb) of the range in which the pore CV is present with respect to a distance t.sub.L from a lowermost portion of the measurement electrode **44** to the lowermost end of the range in which the pore CV is present is in the range of 95:5 to 10:90.

(79) The volume ratio of the solid electrolyte SE in the upper layer **44U** is 20% to 40%, and the volume ratio of the solid electrolyte SE in the lower layer **44L** is 50% to 60%.

(80) In this case, the rate of change of the NOx current Ip2 with respect to an initial value thereof when the gas sensor **100** is used continuously is suppressed to 6% or less. That is to say, a rate of deterioration of measurement sensitivity when the gas sensor **100** is used continuously is suppressed to 6% or less.

(81) The thickness ratio t.sub.U:t.sub.L is more preferably in a range of 90:10 to 30:70. In this case, the rate of change of the NOx current Ip2 with respect to the initial value thereof when the gas sensor **100** is used continuously, that is, the rate of deterioration of measurement sensitivity when the gas sensor **100** is used continuously is suppressed generally to 3% or less. Furthermore, reduction in pumping capacity of the measurement pump cell **41** due to insufficiency of the number of sites for ionization of the measurement gas, which can be caused when a thickness ratio of the upper layer **44U** is small and therefore there are few pores in the measurement electrode **44**, is suppressed.

(82) The configuration of the measurement electrode **44** in which the solid electrolyte contiguous with the first solid electrolyte layer **4** enters into a noble metal portion produces the so-called anchoring effect. The configuration of the measurement electrode **44** used in the gas sensor **100** according to the present embodiment is thus effective at suppressing separation of the measurement electrode **44**.

(83) <Example of Identification of Locations of Boundaries>

(84) A way to identify locations of the respective boundaries (dashed lines La and Lb) when the locations of the boundaries are evaluated will be described next with an example, which are a location of the boundary of the region where the noble metal NM is present and the region where the noble metal NM is not present as the lowermost end of the range in which the noble metal NM is present and a location of the boundary of the region where the pore CV is present and the region where the pore CV is not present as the lowermost end of the range in which the pore CV is present in the thickness direction of the measurement electrode **44** having a configuration as described above.

(85) First, a cross-sectional image of the measurement electrode **44** along the thickness direction as shown in FIG. 2 is captured using a scanning electron microscope (SEM) and the like, for example. In this case, an image capturing range is set so that the measurement electrode **44** is as horizontal as possible in the captured image, and the captured image includes at least the entire range in the thickness direction of the lower layer **44L** of the measurement electrode **44** and a portion near the boundary between the measurement electrode **44** and the first solid electrolyte layer **4**. Image capturing magnification is preferably approximately 500× to 1000× in view of the need to clearly identify each of interfaces among the three phases in the captured image. In this case, the cross-sectional image in a range of approximately 100 μm to 200 μm in the longitudinal direction of the element can be obtained.

(86) Data of the obtained captured image is then analyzed based on a known image processing method to identify the ranges (all the pixels corresponding to the ranges) in which the noble metal NM, the solid electrolyte SE (and the first solid electrolyte layer **4**), and the pore CV (and the third internal space **61**) are present. In a case where the captured image is an SEM image, for example, which phases respective pixels of the captured image correspond to can be identified by a difference in brightness.

(87) A coordinate point (pixel location) providing a location of a lowermost end of the noble metal NM in the captured image is then identified based on the data of the captured image. In a case of

FIG. 2, a point A corresponds to the coordinate point. A straight line passing through the point A and being parallel to a horizontal direction of the captured image is to correspond to the dashed line La showing the lowermost end of the range in which the noble metal NM is present.

(88) However, the measurement electrode **44** might be inclined to no small extent in the captured image, although the measurement electrode **44** is as horizontal as possible at image capturing. In light of the foregoing, the dashed line La showing the lowermost end of the range in which the noble metal NM is present may be set by identifying, from a range of the noble metal NM in the captured image, the coordinate point (e.g., the point A in FIG. 2) of the location of the lowermost end and a coordinate point (e.g., a point B in FIG. 2) at the second lowest location, correcting the captured image so that a straight line passing through the point A and the point B is horizontal, and determining the straight line passing through the point A and the point B in the corrected captured image as the dashed line La.

(89) When the dashed line La is identified, a coordinate point (pixel location) providing a location of a lowermost end of the pore CV in the (corrected) captured image is then identified based on the data of the captured image. In a case of FIG. 2, a point C corresponds to the coordinate point. A straight line passing through the point C and being parallel to the horizontal direction of the captured image is to correspond to the dashed line Lb showing the lowermost end of the range in which the pore CV is present.

(90) A distance between the dashed line La and the dashed line Lb identified as described above is identified as the thickness $t_{\text{sub.L}}$ of the lower layer **44L**.

(91) Furthermore, in FIG. 2, a point D provides an uppermost location of the measurement electrode **44** (upper layer **44U**), and a distance between the dashed line Lb and the point D is identified as the thickness t_{U} of the upper layer **44U**.

(92) <Process of Manufacturing Sensor Element>

(93) A process of manufacturing the sensor element **101** having a configuration and features as described above will be described next. In the present embodiment, a laminated body including green sheets (also referred to as base material tapes) containing zirconia as a ceramic component is formed, and the laminated body is cut and fired to manufacture the sensor element **101**.

(94) Description will be made below by taking, as an example, a case where the sensor element **101** including the six layers shown in FIG. 1 is manufactured. In this case, six green sheets corresponding to the first substrate layer **1**, the second substrate layer **2**, the third substrate layer **3**, the first solid electrolyte layer **4**, the spacer layer **5**, and the second solid electrolyte layer **6** are to be prepared. FIG. 4 is a flowchart of processing at the manufacture of the sensor element **101**.

(95) In a case where the sensor element **101** is manufactured, blank sheets (not illustrated) being green sheets having no pattern formed thereon are prepared first (step S1). In a case where the sensor element **101** including the six layers is manufactured, six blank sheets corresponding to the respective layers are prepared.

(96) The blank sheets have a plurality of sheet holes used for positioning in printing and lamination. The sheet holes are formed to the blank sheets in advance prior to pattern formation through, for example, punching by a punching machine. Green sheets corresponding to layers in which an internal space is formed also include penetrating portions corresponding to the internal space formed in advance through, for example, punching as described above. The blank sheets corresponding to the respective layers of the sensor element **101** are not required to have the same thickness.

(97) After preparation of the blank sheets corresponding to the respective layers, pattern printing and drying are performed on the individual blank sheets (step S2). Specifically, a pattern of various electrodes, a pattern of the heater element **72** and the heater insulating layer **74**, a pattern of internal wiring, which is not illustrated, and the like are formed.

(98) Application or placement of a sublimable material to form the first diffusion control part **11**, the second diffusion control part **13**, and the third diffusion control part **30** is also performed at the

time of pattern printing.

(99) The patterns are printed by applying pastes for pattern formation prepared in accordance with the properties required for respective formation targets onto the blank sheets using known screen printing technology. A known drying means can be used for drying after printing.

(100) A method different from a conventional method is used to form, from among these patterns, a pattern to eventually be the measurement electrode **44**. In this regard, description will be made below.

(101) After pattern printing on each of the blank sheets, printing and drying of a bonding paste are performed to laminate and bond the green sheets corresponding to the respective layers (step S3). The known screen printing technology can be used for printing of the bonding paste, and the known drying means can be used for drying after printing.

(102) The green sheets to which an adhesive has been applied are then stacked in a predetermined order, and the stacked green sheets are crimped under predetermined temperature and pressure conditions to thereby form a laminated body (step S4). Specifically, crimping is performed by stacking and holding the green sheets as a target of lamination on a predetermined lamination jig, which is not illustrated, while positioning the green sheets at the sheet holes, and then heating and pressurizing the green sheets together with the lamination jig using a lamination machine, such as a known hydraulic pressing machine. The pressure, temperature, and time for heating and pressurizing depend on a lamination machine to be used, and these conditions may be determined appropriately to achieve good lamination.

(103) After the laminated body is obtained as described above, the laminated body is cut at a plurality of locations into individual units (referred to as element bodies) of the sensor elements **101** (step S5).

(104) The element bodies obtained by cutting are each fired at a firing temperature of approximately 1300° C. to 1500° C. (step S6). The sensor element **101** is thereby manufactured. That is to say, the sensor element **101** is generated by integrally firing the solid electrolyte layers and the electrodes. The firing temperature in this case is preferably 1200° C. or more and 1500° C. or less (e.g., 1400° C.). Integral firing is performed in this manner, so that the electrodes each have sufficient adhesion strength in the sensor element **101**.

(105) The sensor element **101** thus obtained is housed in a predetermined housing, and built into the body (not illustrated) of the gas sensor **100**.

(106) <Method for Forming Measurement Electrode>

(107) As described above, in the present embodiment, the measurement electrode **44** having a two-layer configuration as described above is provided in the sensor element **101**. Formation of the measurement electrode **44** will be described.

(108) FIG. 5 is a diagram showing procedures for forming the pattern to eventually be the measurement electrode **44**. There are two methods for forming the pattern to be the measurement electrode **44**: a manufacturing method A, and a manufacturing method B.

(109) In the manufacturing method A, an electrolyte paste containing, as a ceramic component, zirconia as with the green sheets is first applied at a location, on an upper surface of a green sheet to form the first solid electrolyte layer **4**, as a target of formation of the measurement electrode **44** (step S21A). The electrolyte paste is a paste obtained by mixing powder of zirconia as the solid electrolyte and an organic component including a binder and the like.

(110) Then, a three-component mixed paste is superimposedly applied onto an applied film formed by application of the electrolyte paste (step S22). The three-component mixed paste is herein a paste obtained by mixing powder of the noble metal containing Pt as the main component, powder of zirconia as the ceramic component, powder of a pore forming material as a sublimable material to form the pore CV in the measurement electrode **44** eventually formed, and an organic component including a binder and the like.

(111) When application of the three-component mixed paste is completed, a paste applied film

(paste-doubled film) where the electrolyte paste and the three-component mixed paste are superimposed is pressed by a predetermined pressing means (step S23). By pressing, some of particles of the noble metal and the pore forming material included in the applied film of the three-component mixed paste enter into the applied film of the electrolyte paste.

(112) After that, when firing targeted for an element body is performed as described above, the paste-doubled film is also fired, and volatilization of the organic component and, further, sintering of the noble metal NM and the solid electrolyte SE progress. In this case, the solid electrolyte in the electrolyte paste is integrated with the solid electrolyte in the green sheet, and a portion in which some particles of the noble metal have entered into the electrolyte paste becomes the lower layer **44L** with the progress of sintering. On the other hand, in a portion in which the three-component mixed paste is applied, the pore forming material sublimates with the progress of sintering of the noble metal NM and the solid electrolyte SE to form the pore CV, and the upper layer **44U** in which the noble metal NM, the solid electrolyte SE, and the pore CV coexist is eventually obtained. As a result, the measurement electrode **44** having a two-layer configuration as shown in FIG. 2 is formed.

(113) Application thicknesses and application areas of the electrolyte paste and the three-component mixed paste are only required to be set by taking into account contraction by firing and entry of the particles of the noble metal and the pore forming material by pressing so that the thicknesses of the upper layer **44U** and the lower layer **44L** of the measurement electrode **44** eventually formed and a planar area of the measurement electrode **44** have desired values. For example, a relationship between the application thicknesses and the application areas of the electrolyte paste and the three-component mixed paste and the thicknesses of the upper layer **44U** and the lower layer **44L** of the measurement electrode **44** and the planar area of the measurement electrode **44** may experimentally be identified in advance. A mixing ratio of the powder of the noble metal, the powder of zirconia as the solid electrolyte, and the powder of the pore forming material of the three-component mixed paste is only required to be set by taking into account entry of the particles of the noble metal and the pore forming material by pressing so that the volume ratios of the noble metal NM, the solid electrolyte SE, and the pore CV in the upper layer **44U** (three-phase region) of the measurement electrode **44** eventually formed have desired values.

(114) On the other hand, in the manufacturing method B, a two-component mixed paste is applied at the location, on the upper surface of the green sheet to form the first solid electrolyte layer **4**, as the target of formation of the measurement electrode **44** (step S21B). The two-component mixed paste is herein a paste obtained by mixing the powder of the noble metal containing Pt as the main component, the powder of zirconia as the ceramic component, and an organic component including a binder and the like.

(115) Hereinafter, the three-component mixed paste is superimposedly applied onto an applied film formed by application of the two-component mixed paste (step S22), and, further, a paste applied film (paste-doubled film) where the two-component mixed paste and the three-component mixed paste are superimposed is pressed by a predetermined pressing means (step S23), as in the manufacturing method A.

(116) Also in a case of the manufacturing method B, some of particles of the noble metal and the pore forming material included in the coating of the three-component mixed paste enter into the coating of the two-component mixed paste by pressing.

(117) After that, when firing targeted for the element body is performed as described above, the paste-doubled film is also fired, and volatilization of the organic component and, further, sintering of the noble metal NM and the solid electrolyte SE progress. A portion in which the two-component mixed paste is applied thus becomes the lower layer. On the other hand, in the portion in which the three-component mixed paste is applied, the pore forming material sublimates with the progress of sintering of the noble metal NM and the solid electrolyte SE to form the pore CV, and the upper layer **44U** in which the noble metal NM, the solid electrolyte SE, and the pore CV coexist

is eventually obtained. Also in this case, as a result, the measurement electrode **44** having a two-layer configuration as shown in FIG. 2 is formed.

(118) Also in the case of the manufacturing method B, the application thicknesses and the application areas of the two-component mixed paste and the three-component mixed paste are only required to be set by taking into account contraction by firing and entry of the particles of the noble metal and the pore forming material by pressing so that the thicknesses of the upper layer **44U** and the lower layer **44L** of the measurement electrode **44** eventually formed and the planar area of the measurement electrode **44** have desired values. A mixing ratio of the powder of the noble metal and the powder of zirconia as the solid electrolyte of the two-component mixed paste is only required to be set by taking into account entry of the particles of the noble metal and the pore forming material by pressing so that the volume ratios of the noble metal NM and the solid electrolyte SE in the lower layer **44L** (two-phase region) of the measurement electrode **44** eventually formed have desired values. Similarly, the mixing ratio of the powder of the noble metal, the powder of zirconia as the solid electrolyte, and the powder of the pore forming material of the three-component mixed paste is only required to be set by taking into account entry of the particles of the noble metal and the pore forming material by pressing so that the volume ratios of the noble metal NM, the solid electrolyte SE, and the pore CV in the upper layer **44U** (three-phase region) of the measurement electrode **44** eventually formed have desired values. As in the case of the manufacturing method A, the relationship among these values may experimentally be identified in advance.

(119) As described above, according to the present embodiment, the measurement electrode of the sensor element of the limiting current type gas sensor is provided to have the two-layer configuration despite that it is provided as the porous cermet electrode as a whole, including the lower layer in which only the noble metal and the solid electrolyte are present and the pore is not present and the upper layer in which the noble metal, the solid electrolyte, and the pore coexist, so that deterioration of the measurement electrode due to the presence of the surplus oxygen not taken into the measurement electrode is suitably suppressed. A gas sensor in which deterioration of measurement sensitivity is suitably suppressed even if the gas sensor is used continuously is thereby achieved.

(120) <Modifications>

(121) The two-layer configuration including the region where the pore is not present and the region where the pore is present used in the measurement electrode **44** in the above-mentioned embodiment may be applied to the inner pump electrode **22** and the auxiliary pump electrode **51** each provided as a cermet of the Au—Pt alloy and ZrO₂ as with the measurement electrode **44**. In this case, a separation suppressive effect associated with the anchoring effect can also be obtained in these electrodes. The inner pump electrode **22** and the auxiliary pump electrode **51** having such a configuration can be formed by a similar process to the measurement electrode **44**.

(122) Although the two methods, that is, the manufacturing method A and the manufacturing method B, are shown in the above-mentioned embodiment as the method for forming the pattern to be the measurement electrode **44**, a manufacturing method C may be used in place of these methods.

(123) In the manufacturing method C, the electrolyte paste used in the manufacturing method A, the two-component mixed paste used in the manufacturing method B, and the three-component mixed paste used in the manufacturing method A and the manufacturing method B are applied in the stated order at the location, on the upper surface of the green sheet to form the first solid electrolyte layer **4**, as the target of formation of the measurement electrode **44**, and then are pressed by a predetermined pressing means.

(124) Also in a case of the manufacturing method C, the application thicknesses and the application areas of the electrolyte paste, the two-component mixed paste, and the three-component mixed paste are only required to be set by taking into account contraction by firing and entry of the particles of the noble metal and the pore forming material by pressing so that the thicknesses of the

upper layer **44U** and the lower layer **44L** of the measurement electrode **44** eventually formed and the planar area of the measurement electrode **44** have desired values. The mixing ratio of the powder of the noble metal and the powder of zirconia as the solid electrolyte of the two-component mixed paste is only required to be set by taking into account entry of the particles of the noble metal and the pore forming material by pressing so that the volume ratios of the noble metal NM and the solid electrolyte SE in the lower layer **44L** (two-phase region) of the measurement electrode **44** eventually formed have desired values. Similarly, the mixing ratio of the powder of the noble metal, the powder of zirconia as the solid electrolyte, and the powder of the pore forming material of the three-component mixed paste is only required to be set by taking into account entry of the particles of the noble metal and the pore forming material by pressing so that the volume ratios of the noble metal NM, the solid electrolyte SE, and the pore CV in the upper layer **44U** (three-phase region) of the measurement electrode **44** eventually formed have desired values. As in the case of the manufacturing method A and the manufacturing method B, the relationship among these values may experimentally be identified in advance.

EXAMPLES

(125) Six types of gas sensors **100** (Examples 1 to 6) differing in conditions for manufacturing the measurement electrode **44** were manufactured as examples, and, for each of the obtained gas sensors **100**, volume ratios of respective phases (the noble metal, the solid electrolyte, and the pore) in each of the upper layer **44U** and the lower layer **44L** of the measurement electrode **44** and the thickness ratio $t_{sub.U}:t_{sub.L}$ of the two layers were evaluated. A continuous drive test in air was also conducted to evaluate a degree of deterioration of the measurement electrode with continued use.

(126) At formation of the measurement electrodes **44**, the measurement electrodes **44** were manufactured by two different manufacturing methods, that is, the manufacturing method A and the manufacturing method B, and, for each of the manufacturing methods, the three-component mixed pastes to mainly form the upper layers **44U** had two different weight ratios of the powder of the noble metal, the powder of the solid electrolyte, and the pore forming material. One type of the electrolyte paste was used in the manufacturing method A, and one type of the two-component mixed paste was used in the manufacturing method B.

(127) A gas sensor including the measurement electrode **44Z** formed using the three-component mixed paste as a whole was also manufactured as a conventional example (a method for manufacturing the gas sensor is hereinafter referred to as a conventional manufacturing method), and, for the manufactured gas sensor, the volume ratios of the respective phases (the noble metal, the solid electrolyte, and the pore) of the measurement electrode **44Z** were evaluated, and the continuous drive test in air was conducted as with the gas sensors of the examples.

(128) Weight ratios (component ratios) of the powder of the noble metal, the powder of the solid electrolyte, and the pore forming material of the pastes used to manufacture the measurement electrodes **44** of Examples 1 to 6 and the measurement electrode **44Z** of the conventional example are shown in Table 1 as a list.

(129) TABLE-US-00001 TABLE 1 MEASUREMENT ELECTRODE PASTE COMPONENT RATIOS (WEIGHT RATIOS) MEASUREMENT FOR UPPER LAYER FORMATION (a:b:c) FOR LOWER LAYER FORMATION (d:e) ELECTRODE NOBLE PORE FORMING NOBLE FORMATION METAL ELECTROLYTE MATERIAL METAL ELECTROLYTE METHOD a b c USED PASTE d e CONVENTIONAL CONVENTIONAL 10 2 1 NOT USED — — EXAMPLE MANUFACTURING METHOD EXAMPLE 1 MANUFACTURING 10 2 1 ELECTROLYTE — 10 METHOD A PASTE EXAMPLE 2 MANUFACTURING 30 10 1 — 10 METHOD A EXAMPLE 3 MANUFACTURING 30 10 1 — 10 METHOD A EXAMPLE 4 MANUFACTURING 10 2 1 TWO- 5 1 METHOD B COMPONENT EXAMPLE 5 MANUFACTURING 30 10 1 MIXED PASTE 5 1 METHOD B EXAMPLE 6 MANUFACTURING 10 2 1 5 1 METHOD B

(130) In Table 1, weight ratios of the three-component mixed pastes are shown in columns “FOR UPPER LAYER FORMATION”, and weight ratios of the electrolyte pastes and weight ratios of the two-component mixed pastes are shown in columns “FOR LOWER LAYER FORMATION”.

(131) When a, b, and c are respectively weight ratios of the powder of the noble metal, the powder of the solid electrolyte, and the pore forming material of the three-component mixed pastes, and d and e are respectively weight ratios of the powder of the noble metal and the powder of the solid electrolyte of the two-component mixed pastes, the weight ratios of the three-component mixed pastes satisfy an equation $a:b:c=10:2:1$ (Examples 1, 4, and 6) or an equation $a:b:c=30:10:1$ (Examples 2, 3, and 5) as shown in Table 1. When the mixing ratio c of the pore forming material of each of the three-component mixed pastes is one, the two-component mixed pastes satisfy an equation $(c):d:e=(1):5:1$ (Examples 4, 5, and 6). Furthermore, as for the weight ratio of the powder of the solid electrolyte of each of the electrolyte pastes, which is expressed as e for the sake of convenience, relative to the mixing ratio c of the pore forming material of each of the three-component mixed pastes, an equation $c:e=1:10$ (Examples 1, 2, and 3) is satisfied.

(132) In each of Examples 1 to 3 in which the measurement electrode **44** is formed by the manufacturing method A, an intended application thickness of the electrolyte paste is 10 μm , and an intended application thickness of the three-component mixed paste is 15 μm .

(133) In each of Examples 4 to 6 in which the measurement electrode **44** is formed by the manufacturing method B, an intended application thickness of the two-component mixed paste is 5 μm , and the intended application thickness of the three-component mixed paste is 15 μm .

(134) In the conventional example in which the measurement electrode **44Z** is manufactured by the conventional manufacturing method, the three-component mixed paste satisfies the equation $a:b:c=10:2:1$, and the intended application thickness of the three-component mixed paste is 15 μm .

(135) The volume ratios of the respective phases in each of the upper layer **44U** and the lower layer **44L** of the measurement electrode **44** and the thickness ratio $t_{\text{sub.U}}:t_{\text{sub.L}}$ of the two layers are shown in Table 2. The volume ratios in the measurement electrode **44Z** of the conventional example are shown in a column “UPPER LAYER” for the sake of convenience.

(136) TABLE-US-00002 TABLE 2 MEASUREMENT ELECTRODE VOLUME RATIOS (%): IN RELATION TO EACH LAYER AS A WHOLE AS 100% THICKNESS RATIO UPPER LAYER (THREE-PHASE REGION) LOWER LAYER (TWO-PHASE REGION) $t_{\text{sub.U}}:t_{\text{sub.L}}$ NOBLE METAL ELECTROLYTE PORE NOBLE METAL ELECTROLYTE UPPER LOWER P1 P2 P3 P4 P5 LAYER $t_{\text{sub.U}}$ LAYER $t_{\text{sub.L}}$ CONVENTIONAL 40 20 40 — — 100 0 EXAMPLE EXAMPLE 1 40 20 40 40 60 90 10 EXAMPLE 2 45 40 15 45 55 80 20 EXAMPLE 3 45 40 15 45 55 95 5 EXAMPLE 4 40 20 40 40 60 30 70 EXAMPLE 5 45 40 15 40 60 70 30 EXAMPLE 6 40 20 40 40 60 10 90

(137) In Table 2, P1(%), P2(%), and P3(%) are respectively the volume ratios of the noble metal NM, the solid electrolyte SE, and the pore CV in the upper layer **44U**, and P4 (%) and P5(%) are respectively the volume ratios of the noble metal NM and the solid electrolyte SE in the lower layer **44L**. An equation $P1+P2+P3=P4+P5=100(\%)$ is satisfied.

(138) The volume ratios of the respective phases in each of the measurement electrodes **44** and the measurement electrode **44Z** were obtained by capturing a cross-sectional SEM image of the electrode, and performing known image processing on the cross-sectional SEM image. The cross-sectional SEM image was generally ternary valued so that the region where the noble metal NM was present was white, the region where the solid electrolyte SE was present was gray, and the region where the pore or the internal space was present was black, and ratios of the areas (ratios of the cross-sectional areas) of the respective regions were set to the volume ratios.

(139) The thickness ratio $t_{\text{sub.U}}:t_{\text{sub.L}}$ of the upper layer **44U** to the lower layer **44L** was obtained by identifying the thickness to of the upper layer **44U** and the thickness $t_{\text{sub.L}}$ of the lower layer **44L** based on the cross-sectional SEM image of each of the measurement electrodes **44**.

(140) The results shown in Table 2 indicate that the measurement electrode **44** has the two-layer

configuration including the upper layer **44U** and the lower layer **44L** in each of Examples 1 to 6. (141) More particularly, values of P1, P2, P3, P4, and P5 are the same in Examples 1, 4, and 6, which differ in the method for manufacturing the measurement electrode **44**, but have in common that the weight ratios of the three-component mixed pastes satisfy the equation $a:b:c=10:2:1$. From among these values, the volume ratio of the noble metal NM in each of the upper layer **44U** and the lower layer **44L** satisfies an equation $P1=P4=40\%$. As for the volume ratio of the solid electrolyte SE, the value of P2 in the upper layer **44U** is 20%, whereas the value of P5 in the lower layer **44L** is 60%, which is greater than the value of P2.

(142) In contrast, in Examples 2, 3, and 5, which differ in the method for manufacturing the measurement electrode **44**, but have in common that the weight ratios of the three-component mixed pastes satisfy the equation $a:b:c=30:10:1$, the values of P1, P2, and P3 are the same, but the values of P4 and P5 in Examples 2 and 3 differ from those in Example 5 by 5%. That is to say, the volume ratio P1 of the noble metal NM in the upper layer **44U** has a value of 45%, and the volume ratio P2 of the solid electrolyte SE in the upper layer **44U** has a value of 40% in each of Examples 2, 3, and 5, but the volume ratio P4 of the noble metal NM in the lower layer **44L** has a value of 45% in each of Examples 2 and 3, and has a value of 40% in Example 5. In response to these values, the volume ratio P5 of the solid electrolyte SE has a value of 55% in each of Examples 2 and 3, and has a value of 60% in Example 5.

(143) In each of the examples, however, the volume ratio of the solid electrolyte SE in the lower layer **44L** is greater than the volume ratio of the solid electrolyte SE in the upper layer **44U**. Specifically, the volume ratio of the solid electrolyte SE in the upper layer **44U** is in a range of 20% to 40%, and the volume ratio of the solid electrolyte SE in the lower layer **44L** is in a range of 50% to 60%.

(144) The volume ratios P1, P2, and P3 of the respective phases in the measurement electrode **44Z** of the conventional example are the same as those of Examples 1 and 4 in each of which the three-component mixed paste having the same weight ratios is used.

(145) As for the thickness ratio $t_{\text{sub.U}}:t_{\text{sub.L}}$ of the upper layer **44U** to the lower layer **44L**, the thickness to has a greater value in each of Examples 1 to 3 manufactured by the manufacturing method A, but the thicknesses to and $t_{\text{sub.L}}$ have a different magnitude relationship in response to the volume ratios in the upper layer **44U** in each of Examples 4 to 6 manufactured by the manufacturing method B.

(146) The thickness ratio $t_{\text{sub.U}}:t_{\text{sub.L}}$, however, is in a range of 95:5 to 10:90 in each of the examples.

(147) The continuous drive test in air was conducted by continuously operating each of the gas sensors in an air atmosphere at $25\pm5^\circ\text{C}$. for 3000 hours, and measuring values of the NOx current (also referred to as NOx output) at the start of operation, after the elapse of 1000 hours, after the elapse of 2000 hours, and after the elapse of 3000 hours. The rate of change of the NOx current at each measurement time to a value of the NOx current at the start of operation was calculated as a NOx output change rate. The NOx output change rate is a value indicative of a degree of deterioration of each of the measurement electrodes **44** and the measurement electrode **44Z**.

(148) FIG. 6 is a graph showing results of the continuous drive test. In FIG. 6, a horizontal axis represents a drive time, and a vertical axis represents the NOx output change rate.

(149) As shown in FIG. 6, in the gas sensor of the conventional example, the NOx output significantly changes over the drive time, and the NOx output change rate taking into account variation is approximately $11.5\pm3.5\%$ after the elapse of 3000 hours, whereas, in the gas sensor of each of Examples 1 to 6, the NOx output change rate remains at most at approximately 6% even after the elapse of 3000 hours. This means that deterioration of the measurement electrode is suppressed even after the elapse of 3000 hours in the gas sensor of each of Examples 1 to 6.

(150) In particular, in the gas sensor of each of examples other than Example 3, the NOx output change rate generally remains at 3% or less after the elapse of 3000 hours, and it is determined that

deterioration of the measurement electrode is more suitably suppressed in the gas sensor.

(151) The above-mentioned results of the continuous drive test indicate that, when the measurement electrode of the sensor element of the limiting current type gas sensor is provided to have the two-layer configuration including the lower layer having the two-phase configuration of the noble metal and the solid electrolyte and the upper layer having the three-phase configuration of the noble metal, the solid electrolyte, and the pore, deterioration of the measurement electrode when the gas sensor is used continuously can be suppressed.

(152) While the invention has been shown and described in detail, the foregoing description is in all aspects illustrative and not restrictive. It is therefore understood that numerous modifications and variations can be devised without departing from the scope of the invention.

Claims

1. A sensor element of a limiting current type NO_x sensor, the sensor element comprising: a base part containing an oxygen-ion conductive solid electrolyte as a constituent material; a measurement internal space into which a measurement gas having an oxygen concentration adjusted in advance is introduced; and a measurement pump cell including a measurement electrode disposed to face the measurement internal space, an out-of-space pump electrode disposed at a location other than the measurement internal space, and a portion of the base part located between the measurement electrode and the out-of-space pump electrode and allowing a pump current proportional to the NO_x concentration of the measurement gas to flow, wherein the measurement electrode includes: an upper layer consisting of a noble metal, the solid electrolyte, and pores; and a lower layer consisting of the noble metal and the solid electrolyte, wherein pores are not present in the lower layer, and a volume ratio of the solid electrolyte in the lower layer to a total volume of the lower layer is greater than a volume ratio of the solid electrolyte in the upper layer to a total volume of the upper layer.

2. The sensor element of the NO_x sensor according to claim 1, wherein the volume ratio of the solid electrolyte in the upper layer is 20% to 40%, and the volume ratio of the solid electrolyte in the lower layer is 50% to 60%.

3. The sensor element of the NO_x sensor according to claim 1, wherein a ratio of a thickness of the upper layer to a thickness of the lower layer is in a range of 95:5 to 10:90.

4. The sensor element of the NO_x sensor according to claim 3, wherein the ratio of the thickness of the upper layer to the thickness of the lower layer is in a range of 90:10 to 30:70.

5. A sensor element of a limiting current type NO_x sensor, the sensor element comprising: a base part containing an oxygen-ion conductive solid electrolyte as a constituent material; a measurement internal space into which a measurement gas having an oxygen concentration adjusted in advance is introduced; and a measurement pump cell including a measurement electrode disposed to face the measurement internal space in a portion of the base part defining the measurement internal space, an out-of-space pump electrode disposed at a location other than the measurement internal space, and a portion of the base part located between the measurement electrode and the out-of-space pump electrode and allowing a pump current proportional to the NO_x concentration of the measurement gas to flow, wherein the measurement electrode includes a noble metal, the solid electrolyte, and pores, a boundary between the base part and the measurement electrode is different from a boundary between a region where the pores are present and a region where the pores are not present in the measurement electrode, and a volume ratio of the solid electrolyte in the region where the pores are not present to a total volume of the region where the pores are not present is greater than a volume ratio of the solid electrolyte in the region where the pores are present to a total volume of the region where the pores are present.

6. The sensor element of the NO_x sensor according to claim 5, wherein a boundary between the base part and the measurement electrode is a lowermost end of a region where the noble metal is

present in the measurement electrode, a boundary between the region where the pores are present and the region where the pores are not present in the measurement electrode is a lowermost end of the region where the pores are present, and a ratio of a distance from an upper end of the measurement electrode to the lowermost end of the region where the pores are present with respect to a distance from the lowermost end of the region where the noble metal is present to the lowermost end of the region where the pores are present is in a range of 95:5 to 10:90.

7. The sensor element of the NOx sensor according to claim 6, wherein the ratio of the distance from the upper end of the measurement electrode to the lowermost end of the region where the pores are present with respect to the distance from the lowermost end of the region where the noble metal is present to the lowermost end of the region where the pores are present is in a range of 90:10 to 30:70.

8. The sensor element of the NOx sensor according to claim 5, wherein the volume ratio of the solid electrolyte in the region where the pores are present is 20% to 40%, and the volume ratio of the solid electrolyte in the region where the pores are not present is 50% to 60%.

9. The sensor element of the NOx sensor according to claim 1, wherein the volume ratio of the solid electrolyte in the upper layer is 20% to 40%, and the volume ratio of the solid electrolyte in the lower layer is 50% to 60%.

10. The sensor element of the NOx sensor according to claim 1, wherein a ratio of a thickness of the upper layer to a thickness of the lower layer is in a range of 95:5 to 10:90.

11. The sensor element of the NOx sensor according to claim 10, wherein the ratio of the thickness of the upper layer to the thickness of the lower layer is in a range of 90:10 to 30:70.

12. The sensor element of the NOx sensor according to claim 2, wherein a ratio of a thickness of the upper layer to a thickness of the lower layer is in a range of 95:5 to 10:90.

13. The sensor element of the NOx sensor according to claim 12, wherein the ratio of the thickness of the upper layer to the thickness of the lower layer is in a range of 90:10 to 30:70.

14. The sensor element of the NOx sensor according to claim 9, wherein a ratio of a thickness of the upper layer to a thickness of the lower layer is in a range of 95:5 to 10:90.

15. The sensor element of the NOx sensor according to claim 14, wherein the ratio of the thickness of the upper layer to the thickness of the lower layer is in a range of 90:10 to 30:70.

16. The sensor element of the NOx sensor according to claim 6, wherein the volume ratio of the solid electrolyte in the region where the pores are present is 20% to 40%, and the volume ratio of the solid electrolyte in the region where the pores are not present is 50% to 60%.
